TTM: Training-free Temporal Token Merging for Memory-Efficient Video 3D Reconstruction

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Abstract

Recent video-to-3D foundation models such as VGGT achieve state-of-the-art reconstruction by jointly attending to all *input frames, but the resulting global attention scales quadratically with sequence length and exhausts a 16 GB GPU at roughly thirty frames. We present* ***TTM*** *(Temporal Token Merging), a fully* training-free *add-on that compresses cross-frame attention through three complementary techniques.* ***(M1) Layer-Selective Storage*** *retains full intermediate features only at the four layers tapped by the prediction heads, eliminating roughly 80% of stored activations.* ***(M2) KV-Asymmetric Merge*** *routes full-resolution queries against merged keys/values; this avoids an error-prone unmerge step and structurally prevents token contamination.* ***(M5) Epipolar-Constrained Merge*** *multiplies the bipartite similarity by an exponential decay over the epipolar residual, where the fundamental matrix is computed from camera poses estimated in a fast low-resolution pre-pass. On a single 16 GB GPU, TTM extends the feasible video length from 30 to over 100 frames while keeping the per-pixel depth correlation above* 0.9999 *relative to vanilla VGGT. All components reuse the frozen VGGT weights, require no fine-tuning, and stack with existing efficient-attention kernels.*

1. Introduction

Reconstructing the 3D structure of an environment from ordinary RGB video is a long-standing computer-vision problem with immediate consequences for AR/VR, robotics, and autonomous driving. The recent VGGT model [**?**] demonstrates that a single transformer trained on multi-view supervision can recover per-pixel depth, dense point maps, and camera poses for arbitrarily many input views in a single forward pass, surpassing classical multi-view stereo and prior feed-forward reconstructors [**? ?**]. Its main ingredient is an *alternating attention* mechanism that interleaves per-frame self-attention with a fully global attention block over all $S \cdot P$ tokens in the sequence.

직관 (Intuition)

S 프레임짜리 영상을 한꺼번에 보고 3D를 만드는 AI가 있습니다. 하지만 영상이 길어지면 본부 책상이 무너지듯 GPU 메모리가 폭발합니다. TTM은 "앞 프레임과 거의 같은 부분은 한 장만 쓰자"라는 단순한 발상을 추론 단계에 끼워넣어, 같은 GPU로 2∼3배 긴 영상을 처리할 수 있게 합니다.

The memory wall. The global attention layer requires forming an $(SP) \times (SP)$ score matrix; for VGGT-1B at $S = 20$ this already amounts to roughly $10{,}000 \times 10{,}000$ entries, and the activations of the 24 alternating blocks dominate the memory budget. Concretely, on an NVIDIA RTX 4080 (16 GB), vanilla VGGT runs to completion at $S = 20$ but out-of-memory at $S \geq 30$. Yet many practical sources of 3D supervision—driving footage, head-mounted cameras, drone flights—naturally produce sequences of hundreds of frames.

Why temporal redundancy is special. Adjacent frames in a 30 fps video differ by a 33 ms time step; their tokenised representations are nearly identical apart from small parallax. Single-image token merging methods such as ToMe [**?**] exploit *spatial* redundancy *within* a single image, leaving the much larger *temporal* redundancy across frames untapped. Naively applying ToMe to the stacked token sequence ignores the geometric structure that ties adjacent frames together—and, as we show in Sec. 3, also corrupts the features that downstream prediction heads depend on.

Contributions. We propose TTM with three complementary, training-free modules:

- **C1.** A cross-frame bipartite token-merging scheme with a Δt-aware merge ratio, parameter-free aside from $r_{\max}$ and a decay rate λ.
- **C2.** *Layer-Selective Storage (M1)* — we observe that VGGT's DPT-style heads tap only four of the 24 layers ($\mathcal{T} = \{4, 11, 17, 23\}$). Storing `None` placeholders elsewhere frees ∼870 MB without any output change.

- **C3.** *KV-Asymmetric Merge (M2)* — running global attention with full queries but merged keys/values eliminates the unmerge step and structurally prevents the token contamination caused by averaging across frames.
- **C4.** *Epipolar-Constrained Merge (M5)* — a geometric prior obtained by a low-resolution self-distillation pass. We compute the inter-frame fundamental matrix and multiply the cosine-similarity matrix by $\exp(-d_{\text{epi}}^2/\sigma^2)$, costing $\sim$10% extra wall time but unlocking aggressive merging at $r_{\max} = 0.5$ without quality loss.

On our benchmark (`pyramid.mp4`, RTX 4080 16 GB), TTM fits $S = 60$ at 13.21 GB (vanilla OOMs at $S = 30$) and extends to $S = 100$ via mode and resolution selection, all while preserving a depth correlation of ≥ 0.9999 relative to the un-merged baseline. The full sweep is reported in Sec. 5 (omitted in this draft).

2. Related Work

Holistic video-to-3D models. DUSt3R [**?**] and MASt3R [**?**] pioneered the idea that a single transformer could jointly output dense correspondences and 3D pointmaps from a pair of images. VGGT [**?**] extended the approach to arbitrarily many views via alternating frame/global attention and became the CVPR 2025 best paper. Their strength—joint reasoning across the full token set—is also their weakness: memory scales as $\mathcal{O}((SP)^2)$ and longer videos become infeasible.

Token reduction in vision transformers. ToMe [**?**] introduces a bipartite soft-matching that merges similar tokens within a single image; DiffRate [**?**] learns the merge ratio jointly with the model, and EViT [**?**] drops uninformative tokens. All of these are *spatial*: they operate within one image and assume no inter-image structure. Lite-VGGT [**?**] is the only contemporary work we are aware of that targets VGGT-style models, but it requires re-training and does not exploit geometry. TTM, in contrast, is plug-in, geometry-aware, and operates across the temporal axis.

Memory-efficient attention. FlashAttention [**?**] reduces peak activations during the attention computation itself, Linformer [**?**] projects keys/values to a fixed low-rank subspace, and sparse-attention variants restrict the set of attended tokens to a pattern. These methods are orthogonal to TTM: TTM reduces the *number* of tokens entering attention, which compounds multiplicatively with any of the above.

Geometric priors in transformers. Epipolar attention has been used in multi-view stereo networks [**?**] and pose-aware transformers [**?**] to restrict the attended locations to geometrically consistent pairs, but these methods typically require ground-truth camera poses at training time. Our M5 module instead obtains poses cheaply at inference time via a low-resolution forward pass through the same frozen VGGT, turning the model into its own pose oracle.

3. Method

3.1. Preliminaries: VGGT in one paragraph

Given S RGB frames each tokenised by a patch embedder of patch size 14, VGGT prepends 5 special tokens per frame (1 camera + 4 register), giving $P = 5 + P_{\text{patch}}$ tokens per frame and a total of SP tokens per batch element. A stack of $L = 24$ alternating blocks then repeatedly applies *(i)* frame attention—self-attention within each frame operating on P tokens—and *(ii)* global attention—a full attention over all SP tokens. We denote the ℓ-th layer's output by $Z_\ell \in \mathbb{R}^{1\times S\times P\times 2C}$; the prediction heads (depth, pointmap, camera) consume the list $\{Z_\ell\}_{\ell=1}^{L}$ but only *actually tap* the four layers $\mathcal{T} = \{4, 11, 17, 23\}$.

3.2. Cross-Frame Bipartite Token Merging

For each pair of adjacent frames $(t, t{+}1)$ we project the patch tokens through VGGT's *frozen* key projection $W_K \in \mathbb{R}^{C\times C}$ to obtain $\mathbf{K}_t, \mathbf{K}_{t+1} \in \mathbb{R}^{P_{\text{patch}}\times C}$ (special tokens are excluded). After L2-normalisation, we split the rows of $\mathbf{K}_t$ into source (even indices) and destination (odd indices) sets [**?**], and for each source patch i we find its best match in $\mathbf{K}_{t+1}$:

$$j^\star(i) = \arg\max_j \langle \mathbf{K}_t[i],\, \mathbf{K}_{t+1}[j] \rangle. \tag{1}$$

The top-n source patches by similarity score are then merged with their matches, where the merge count n is determined by a Δt-aware ratio:

$$n = \lfloor P_{\text{patch}} \cdot r(\Delta t) \rfloor, \quad r(\Delta t) = r_{\max} \exp(-\lambda \Delta t_{\text{norm}}). \tag{2}$$

Smaller frame intervals admit more merging; for uniformly sampled videos $\Delta t_{\text{norm}} = 1$ for all pairs and the ratio collapses to a constant.

Mix mode matters. The merged token can either average the two source/destination features (*soft*) or retain the source frame's content verbatim (*hard_kept*). We empirically find that *soft* averaging propagates information from frame $t{+}1$ backward into the slot belonging to frame t, which the position-aware downstream heads interpret as inconsistent evidence and produce blurred depth maps. *hard_kept* removes the contamination entirely, at the cost of throwing away whatever extra information frame $t{+}1$ might offer at that pixel.

직관 (Intuition)

"두 프레임에서 가장 닮은 토큰끼리 짝지어 하나로 합치자." SP 토큰이 $S(P-n)$로 줄어들어 global attention의 $\mathcal{O}((SP)^2)$ 비용이 그만큼 작아집니다. 하지만 "합친다"라는 단순한 평균이 후속 단계에 작은 독을 남길 수 있어, 다음 절들에서 그 독을 어떻게 차단할지 다룹니다.

3.3. M1 | Layer-Selective Storage

A direct profile of VGGT-1B at $S = 20$ reveals that storing the 24 intermediate tensors $\{Z_\ell\}$ accounts for roughly 1 GB of memory. Yet only the four layers in $\mathcal{T}$ are ever consumed downstream, so the remaining 20 tensors are dead weight. M1 simply stores `None` for $\ell \notin \mathcal{T}$; the camera head, which reads only the last element of the list, is unaffected since $23 \in \mathcal{T}$. The change is one `if` statement in the aggregator wrapper and saves ~ 870 MB at $S = 20$, growing approximately linearly in S.

3.4. M2 | KV-Asymmetric Merge

The naive way to apply temporal merging is the *merge-unmerge* pattern of Fig. **??** (top):

1. merge: $X \in \mathbb{R}^{SP\times C} \to X' \in \mathbb{R}^{S(P-n)\times C}$,
2. global attention on X',
3. unmerge: $X' \to Y \in \mathbb{R}^{SP\times C}$ by broadcasting averaged features back to the dropped source positions.

Step 3 is the culprit: after unmerge the dropped source position contains a feature that has *both* frame t and frame $t{+}1$ baked in, breaking the alignment between content and positional embedding that the heads were trained with. We observe this as low-frequency depth bias and lost fine detail.

The asymmetric remedy. Standard attention computes Q, K, V from the *same* token sequence. We instead compute Q from the *full* sequence and K, V from the *merged* sequence:

$$\mathrm{Attn}(Q_{\text{full}}, K_{\text{merged}}, V_{\text{merged}}) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V, \quad (3)$$

where the resulting matrix lives in $\mathbb{R}^{SP\times C}$ already. No unmerge step is required because Q's shape dictates the output shape. Concretely, we obtain Q, K, V by slicing the frozen $W_{qkv} \in \mathbb{R}^{3C\times C}$ into three blocks and applying W_Q to the un-merged tokens and W_K, W_V to the merged tokens; rotary positional embeddings are applied accordingly. The remainder of the block (`proj`, MLP, residuals, LayerNorms) is identical to the original VGGT block.

직관 (Intuition)

"내 작업물 Q는 그대로 두고, 참고자료 K, V만 압축본을 본다." 복사 · 붙여넣기 단계가 사라져 토큰 오염이 원천 차단되며, 출력 텐서가 자연스럽게 원래 크기를 회복합니다.

3.5. M5 | Epipolar-Constrained Merge

Core formula. Sections 3.2–3.4 reduce memory but the matching itself is still purely *semantic*: we pair tokens whose key vectors are similar. In videos, however, there is a strong additional cue—two tokens that depict the same 3D point in adjacent frames must lie on each other's epipolar lines. M5 turns this into a multiplicative gate on the similarity matrix:

$$s_{ij} \;=\; s_{ij}^{\cos} \cdot \exp\!\left(-\frac{d_{\text{epi}}(i,j)^2}{\sigma^2}\right), \quad (4)$$

where $d_{\text{epi}}(i,j)$ is the Euclidean distance, in pixels, between the centre of patch j in frame $t{+}1$ and the epipolar line that patch i in frame t induces.

The chicken-and-egg problem. Computing d_{epi} requires the inter-frame fundamental matrix $\mathbf{F}_{t,t+1}$, which in turn requires camera poses. But VGGT's camera head only produces poses from the *final-layer* aggregated tokens, while our merges happen *inside* the alternating attention stack. We resolve the circular dependency with a low-resolution prepass.

Two-pass self-distillation. *Pass 0* runs the full TTM pipeline (with M1, M2, no M5) at a small target side of 224 px. Because the input is small and global attention is the bottleneck, Pass 0 costs roughly one-tenth the time of the main pass. From Pass 0 we extract $\{\mathbf{R}_t, \mathbf{t}_t\}_{t=1}^{S}$ and per-frame intrinsics $\{K_t\}$. We then rescale the intrinsics to the high-resolution image size (H, W) (the camera is unchanged; only the pixel grid is). The fundamental matrix between two adjacent frames is

$$\mathbf{F}_{t,t+1} \;=\; K_{t+1}^{-\top}\,[\mathbf{t}_{\text{rel}}]_\times\,\mathbf{R}_{\text{rel}}\,K_t^{-1}, \quad (5)$$

with $\mathbf{R}_{\text{rel}} = \mathbf{R}_{t+1}\mathbf{R}_t^\top$ and $\mathbf{t}_{\text{rel}} = \mathbf{t}_{t+1} - \mathbf{R}_{\text{rel}}\mathbf{t}_t$. If the relative pose is degenerate ($\|\mathbf{t}_{\text{rel}}\| < \varepsilon$ and $\|\mathbf{R}_{\text{rel}} - I\| < \varepsilon$) we set $\mathbf{F} = 0$ so that the gate falls back to a uniform 1 and M5 reduces to plain TTM.

Pass 1 repeats the forward at the user-requested resolution (default 518), injecting the precomputed $S-1$ epipolar gates into every merge step. Because the gate is purely geometric and does not depend on the layer index ℓ, the same set of gates is shared across all 20 merge layers, costing only $4(S{-}1)P_{\text{patch}}^2$ bytes ($\sim$440 MB at $S = 60$, $P_{\text{patch}} = 1369$).

직관 (Intuition)

"같은 3D 점이 두 시점 사진에서 어디에 찍히는가"라는 기하학적 사실을 이용해, 비슷해 보이지만 사실 다른 통로의 사과끼리 잘못 묶는 일을 막습니다. 카메라 위치는 같은 모델을 한 번 더 빠르게 돌려서 얻습니다 ($\approx$10% 추가 시간).

Sigma choice and sanity. We use $\sigma = 0.05\,W$ (5% of the image width) as the default; the gate decays to e^{-1} at one σ. Importantly, as $\sigma \to \infty$ the gate becomes the all-ones matrix and Eq. (4) reduces exactly to the M1+M2 baseline. We use this property as a code-level sanity check: enabling M5 with $\sigma = 10^9$ at $S = 10$ produces a numerically identical output (depth L1 $= 0.0000$, correlation $= 1.0000$) to running TTM(M1+M2) directly.

3.6. Implementation Details

No fine-tuning. All three modules operate on frozen VGGT-1B weights. The only learnable parameters are those already present in the released checkpoint; TTM adds only hooks around the alternating-attention loop.

Computational overhead. M1 reduces both memory and time. M2 keeps the attention matmul size at $SP \times S(P-n)$ instead of $SP \times SP$, so it strictly reduces compute. M5 adds a small Pass 0 ($\sim$10% wall time) and a $\mathcal{O}((S-1)P^2_{\text{patch}})$ pre-computation on CPU.

Default hyperparameters. $r_{\max} = 0.3$ for M1+M2 and 0.5 for M1+M2+M5; $\lambda = 3.0$; $\sigma = 0.05\,W$; first merging layer index $\ell_{\min} = 5$ (to protect the fine-grained edges produced by the first DPT tap at layer 4). Special tokens are never merged.

Table 1. Quantitative comparison on the Tanks and Temples training set (5 scenes). Reconstruction quality: F1 score at $\times 5$ and $\times 10$ multiples of the scene-specific d_τ threshold. Pose quality: AUC at 30° (fraction of frames with pose error $< 30°$). Inference time: forward-pass wall-clock time on a single H100 GPU (BF16 for all methods). **Bold**: best per metric. Underline: second best.

Scene	Method	Reconstruction F1 (%) ↑		Pose	Efficiency	
		$\times 5$	$\times 10$	AUC@30° (%) ↑	Time (s) ↓	Mem (GB) ↓
Barn	Vanilla VGGT	48.1	72.7	**91.2**	319.5	83.7
	TTM (Ours)	**50.1**	**74.0**	**91.2**	41.2	38.5
	TTM, w/o epi.	42.0	70.8	89.0	29.6	35.8
	LiteVGGT	28.1	50.2	89.2	**10.7**	**17.9**
Caterpillar	Vanilla VGGT	32.6	50.0	**93.5**	279.5	78.5
	TTM (Ours)	33.0	50.3	93.2	36.5	36.4
	TTM, w/o epi.	28.1	46.8	91.6	26.3	33.9
	LiteVGGT	**35.9**	**53.3**	91.8	**9.9**	**16.7**
Ignatius	Vanilla VGGT	66.7	78.1	98.1	132.5	55.5
	TTM (Ours)	**67.2**	**78.5**	98.1	19.2	27.1
	TTM, w/o epi.	51.6	66.4	97.0	13.7	25.4
	LiteVGGT	31.3	52.6	**99.2**	**7.1**	**12.2**
Meetingroom	Vanilla VGGT	**58.1**	72.4	**97.3**	262.1	76.2
	TTM (Ours)	57.8	72.1	97.0	34.8	35.5
	TTM, w/o epi.	57.6	**74.0**	95.7	24.7	33.0
	LiteVGGT	44.4	65.0	92.9	**9.7**	**16.4**
Truck	Vanilla VGGT	**69.1**	**84.4**	**99.6**	120.9	53.2
	TTM (Ours)	**69.1**	84.3	**99.6**	17.8	26.2
	TTM, w/o epi.	59.9	80.2	99.2	12.7	24.5
	LiteVGGT	39.6	60.4	99.2	**7.0**	**11.9**
4**Average*	Vanilla VGGT	54.9	71.5	**95.9**	222.9	69.4
	TTM (Ours)	**55.4**	**71.8**	95.8	29.9	32.7
	TTM, w/o epi.	47.8	67.6	94.5	21.4	30.5
	LiteVGGT	35.9	56.3	94.4	**8.9**	**15.0**